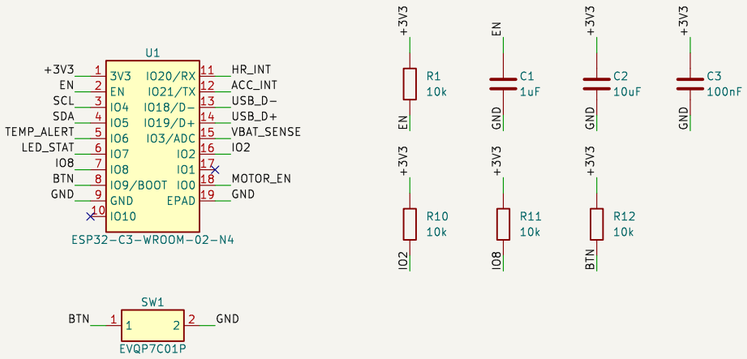
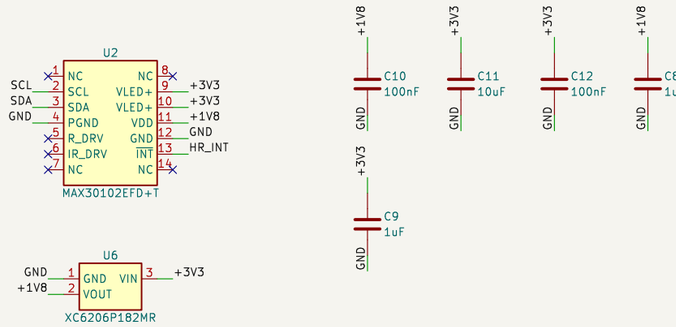
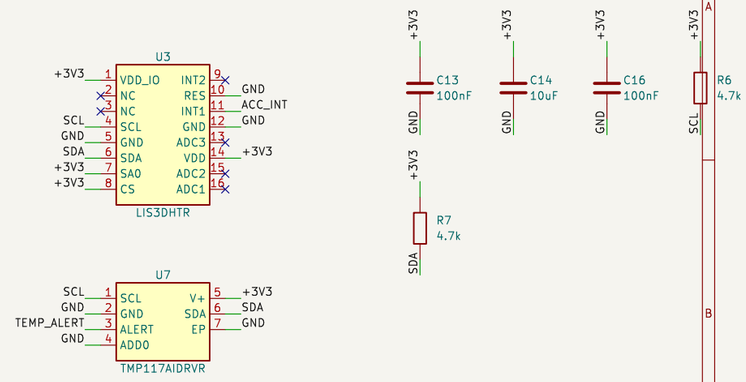
MCU ESP32-C3 (WiFi + BLE, native USB)



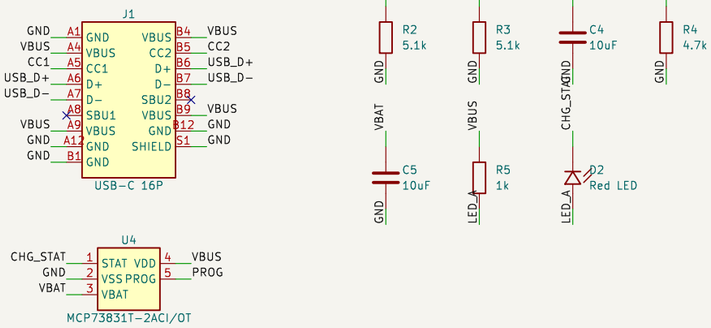
Heart rate + SpO2 (MAX30102, skin side)



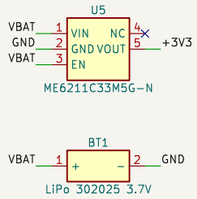
Steps + skin temperature (LIS3DH, TMP117)



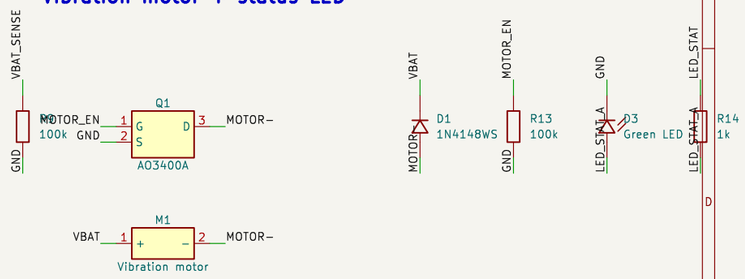
USB-C + Li-Po charger



3.3 V supply + battery sense



Vibration motor + status LED



SPITALUL VIRTUAL BAND (screenless wearable) — made by Stancu Patrick

Every pin has a net label: same label text = same wire. Generated from scripts/design.py.

Power: USB-C 5V -> MCP73831 charger (~210 mA) -> Li-Po -> ME6211 3.3V. MAX30102 core runs from XC6206 1.8V.

I2C bus: MAX30102 0x57, LIS3DH 0x19 (SA0 high), TMP117 0x48 (ADD0 low). TMP117 faces the skin through its own window.

Hobby project — heart-rate / SpO2 / temperature readings are NOT for medical use.

ESP32-C3 + MAX30102 + LIS3DH + TMP117, USB-C charging, no screen

made by Stancu Patrick

Sheet: /

File: spitalul_virtual_band.kicad_sch

Title: Spitalul Virtual Band

Size: A3	Date: 2026-10-27	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1